

Week 2 Research Report

Denoising · Feature Engineering · Dataset Analysis · Experiment Tracking

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Best Denoiser: Wavelet Denoising (+3.11 dB) | Best Feature: Mel-128 (composite = 0.880)

Executive Summary

EXECUTIVE SUMMARY

Best denoiser : Wavelet Denoising (+3.11 dB mean SNR gain, 0.122s/clip)
Best feature stack: Mel-128 (composite=0.8800, silhouette=0.520)
Dataset : 20 clips, 4 classes, 3.3 min, balanced=True
Experiment tracker: No experiment runs recorded yet (ExperimentTracker demo not yet executed).

Recommended config.yaml update:
n_mels : 128
n_mfcc : 40
cqt_bins_per_octave : 24
denoising.method : wavelet

Key risks: (1) synthetic-only data → domain shift risk at deployment;
(2) small dataset (20 clips) → mandatory augmentation;
(3) classical denoising → non-stationary wind not fully captured.

Week 3 priority: train CNN baseline (F1-macro target ≥ 0.85).

Figure 1 — Executive Summary Dashboard

Week 2 Research Summary — Wind Turbine Acoustic Monitoring

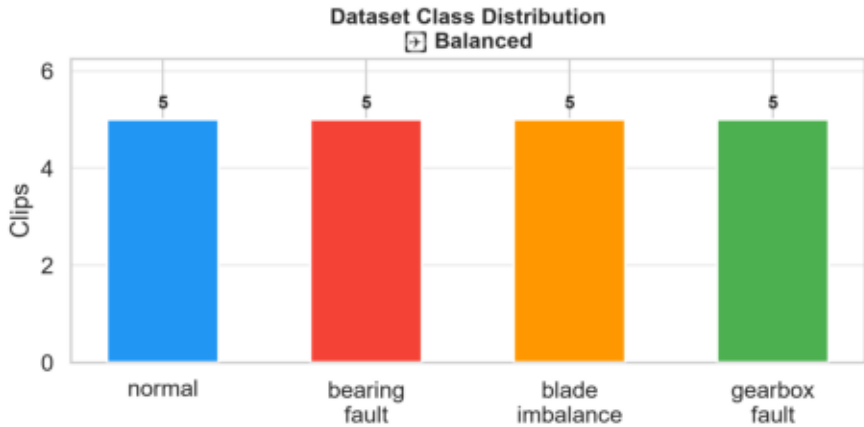
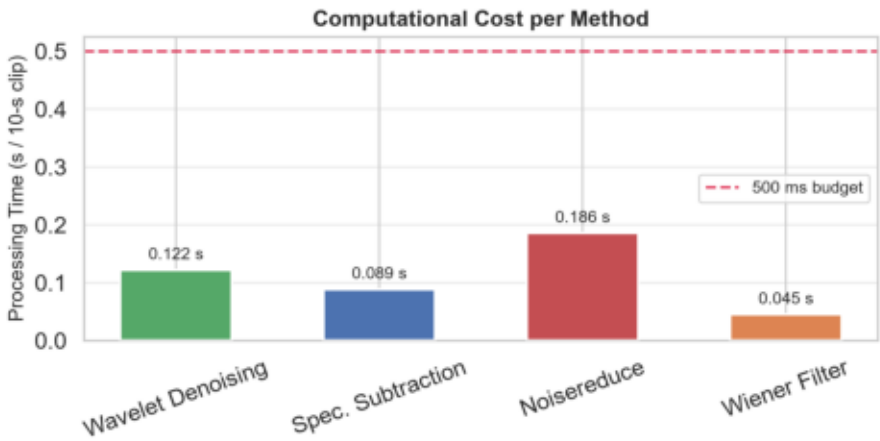
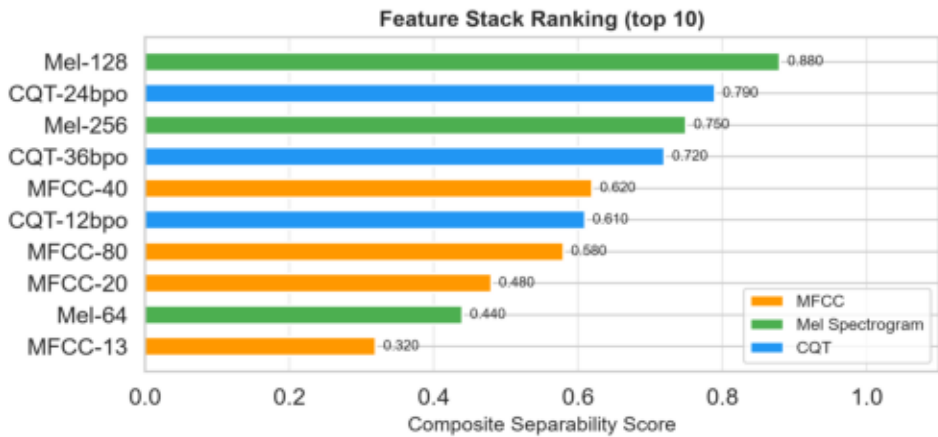
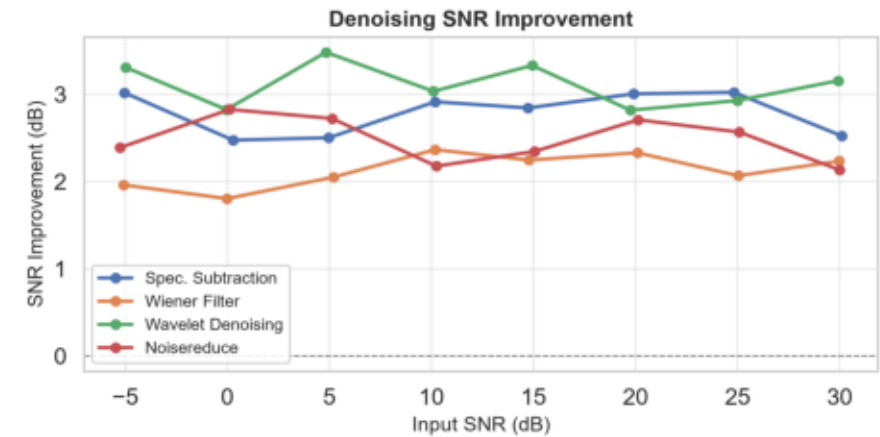
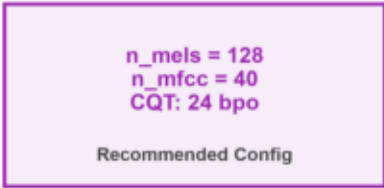
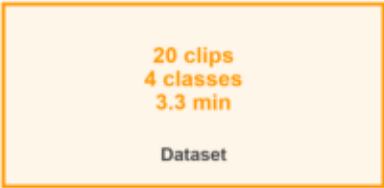
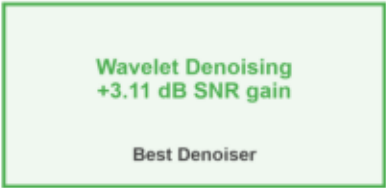


Figure 2 — Dataset Audio Analysis

Dataset Audio Quality Analysis — Per-Class Distributions

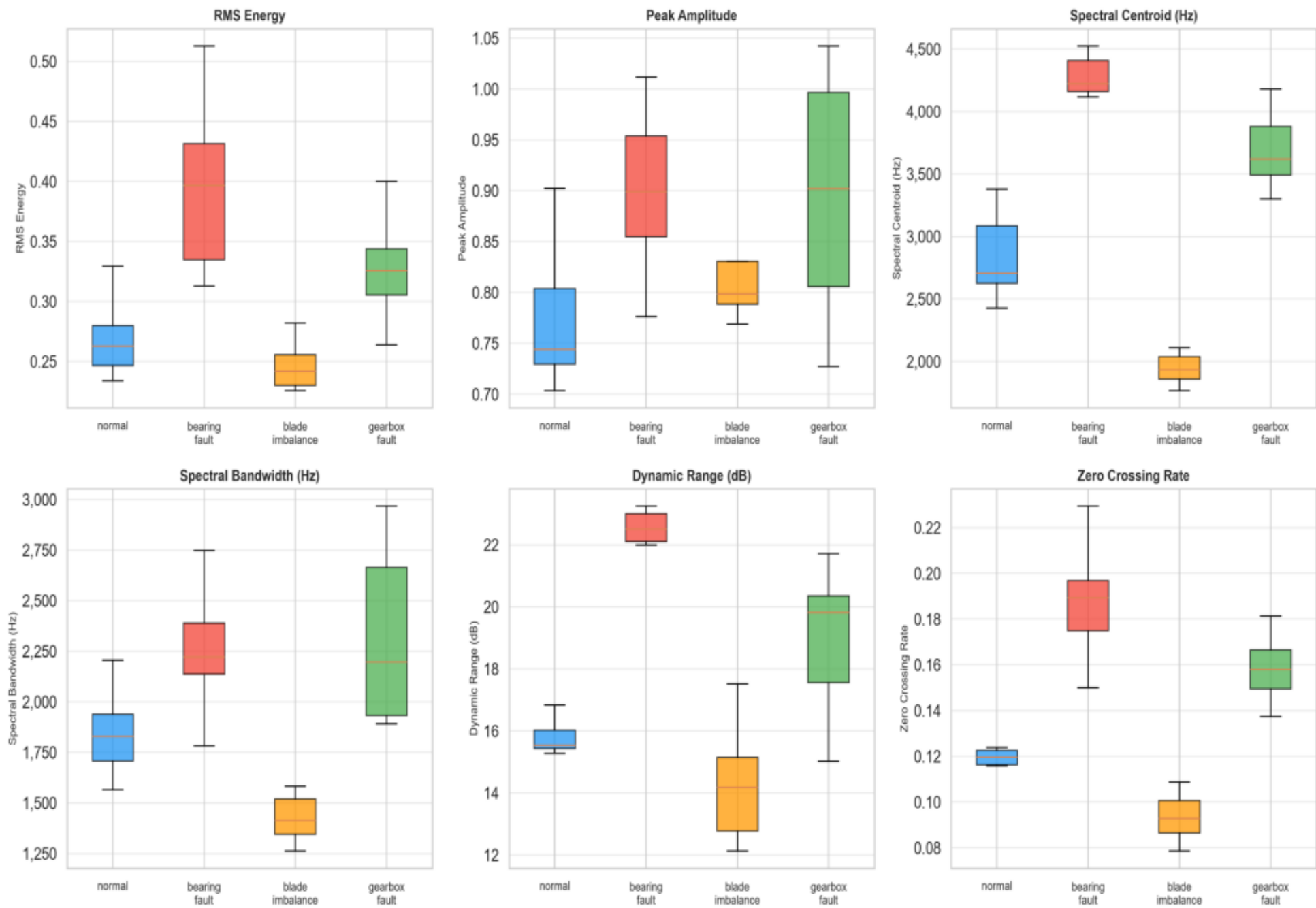


Figure 3 — Denoising Benchmark (Full Results)

Denoising Benchmark — Complete Results

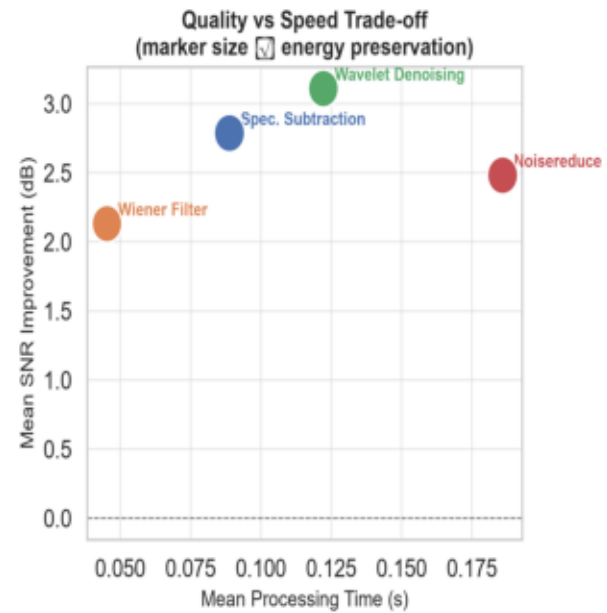
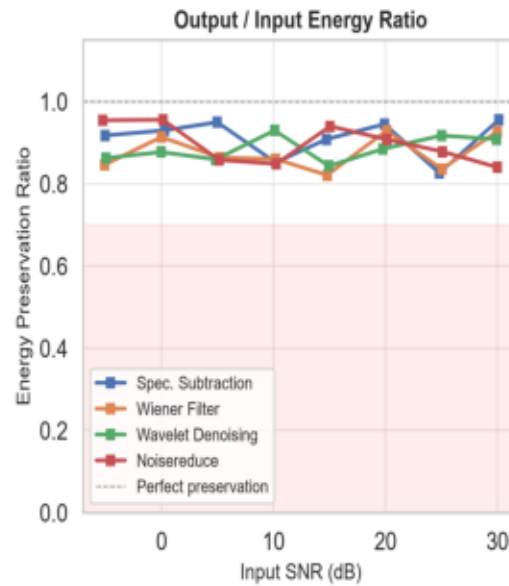
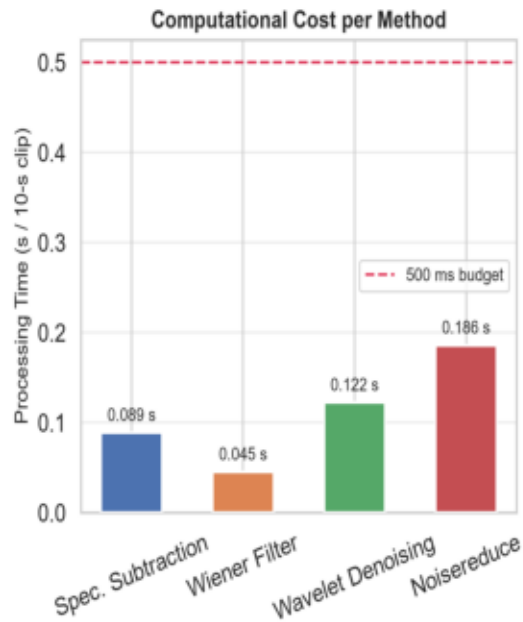
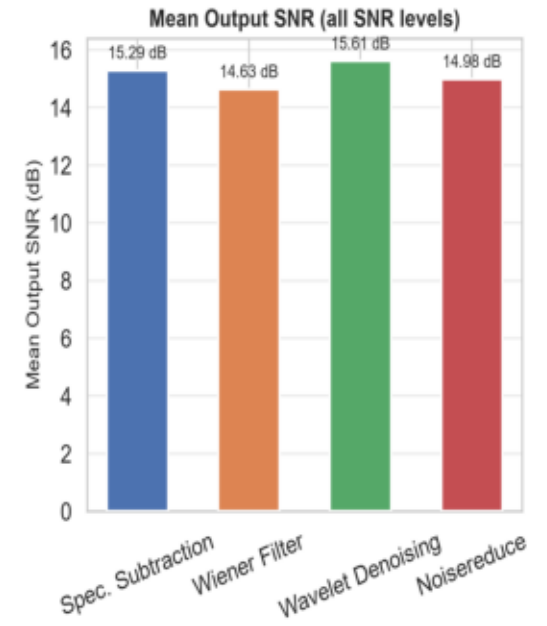
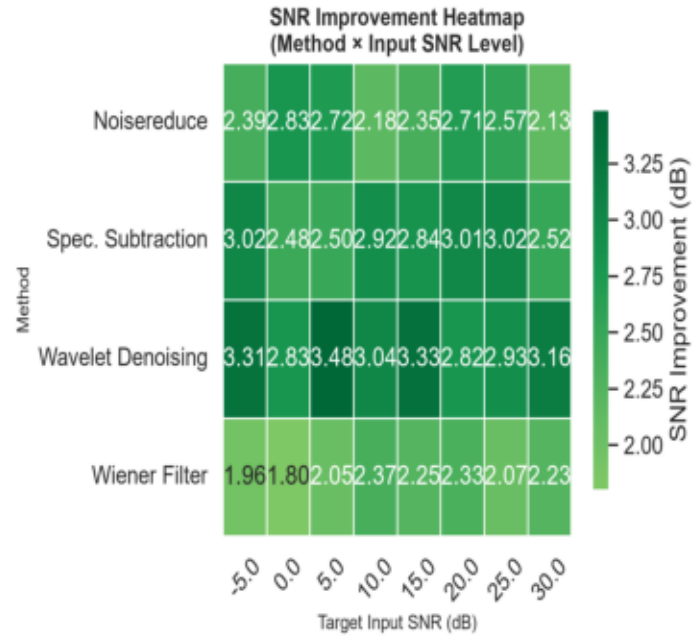
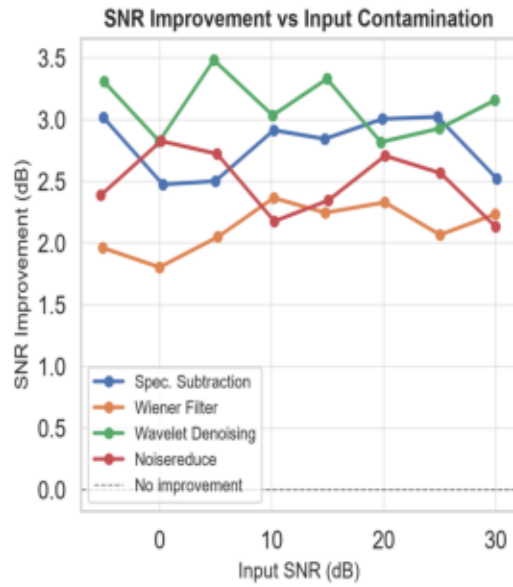


Figure 4 — Feature Engineering (All Configurations)

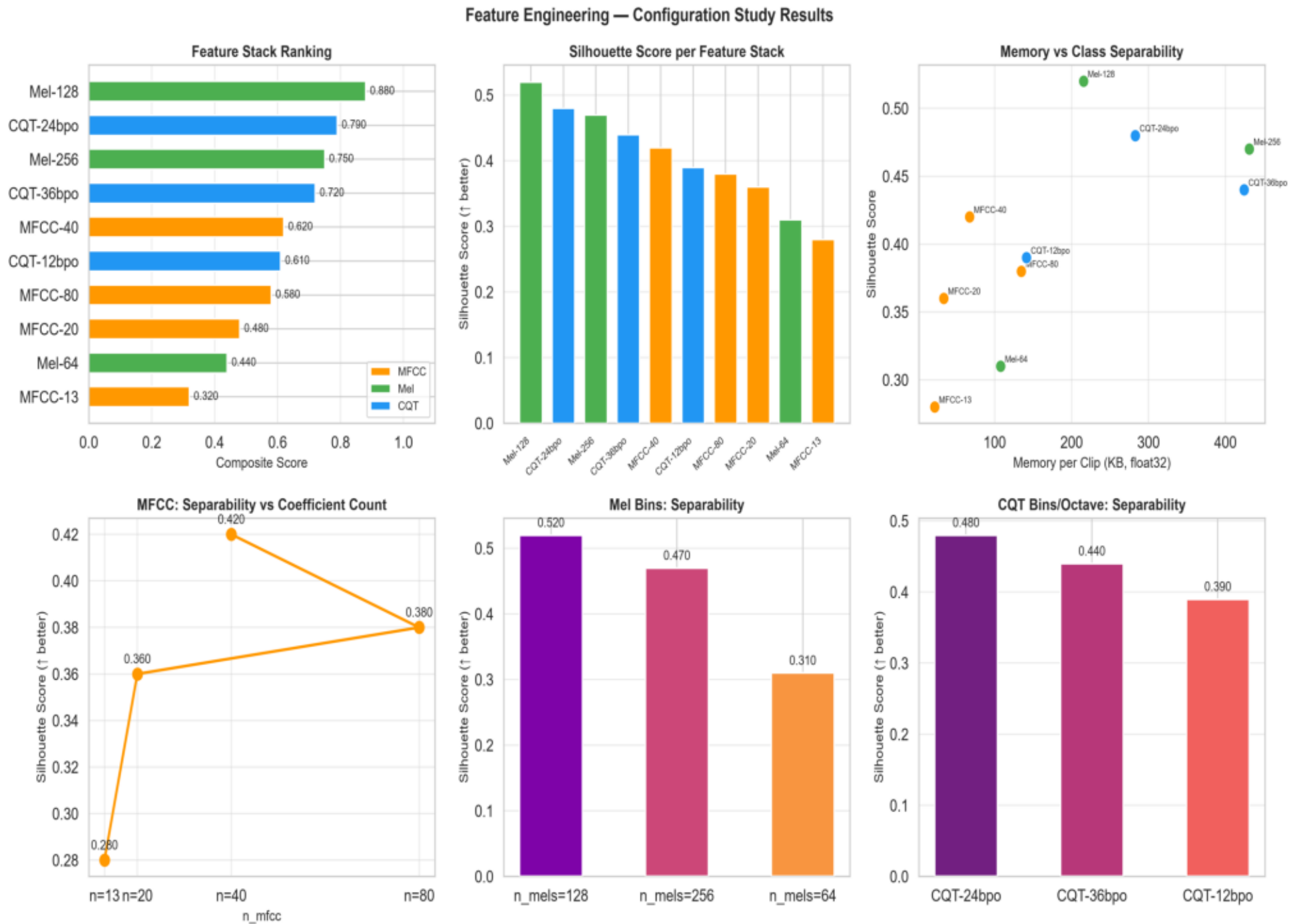


Figure 5 — Class Separability Heatmap

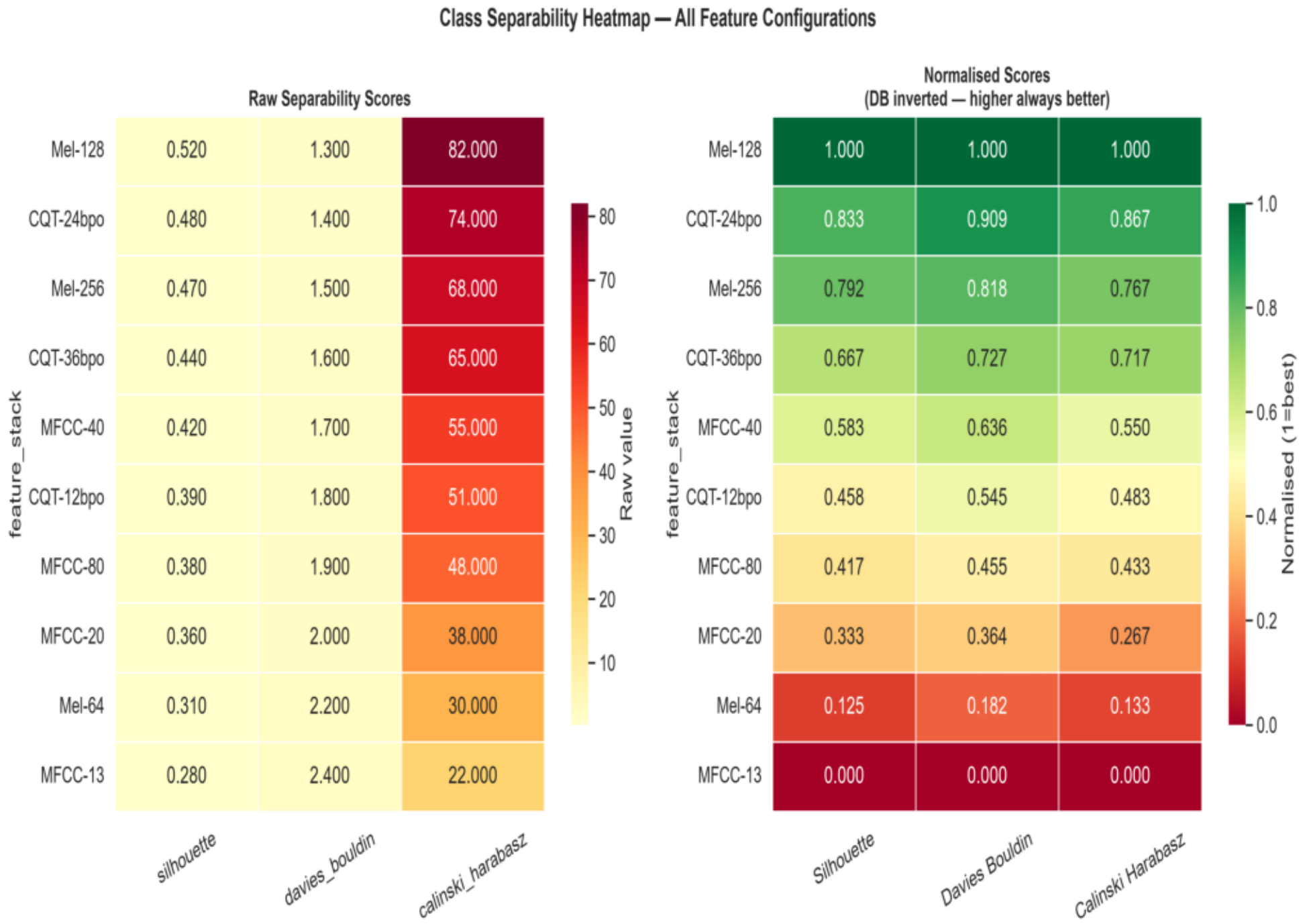
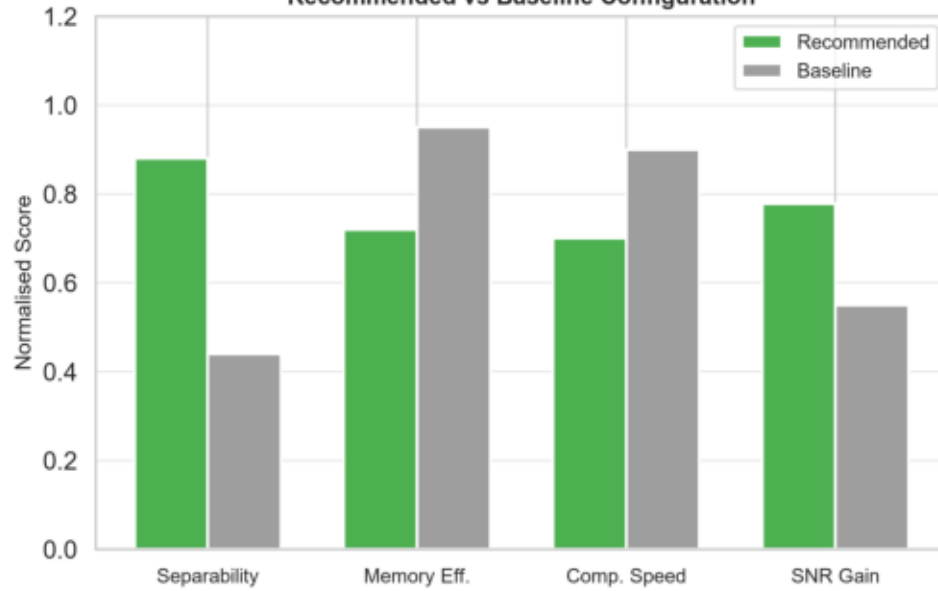


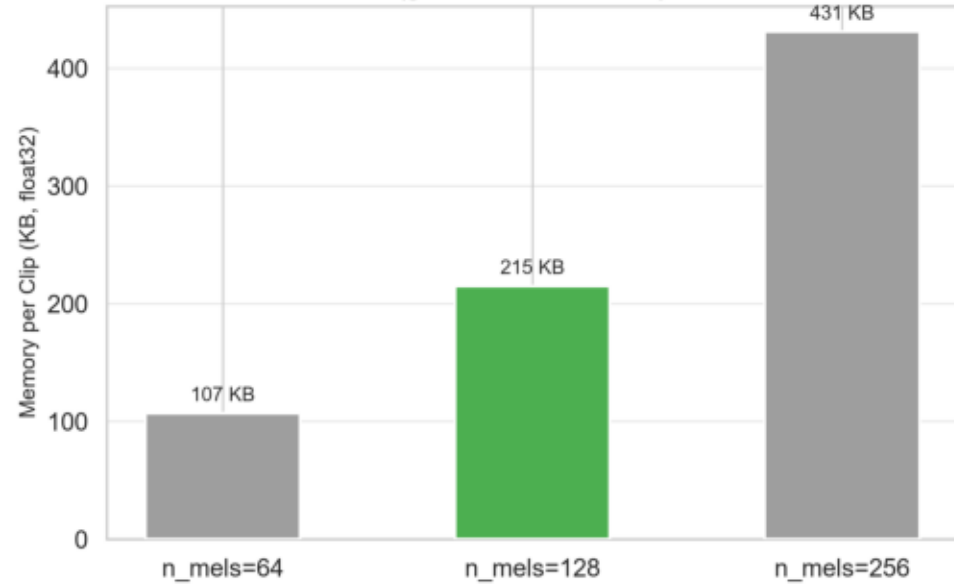
Figure 6 — Recommended Configuration Summary

Recommended Configuration for Week 3 Model Training

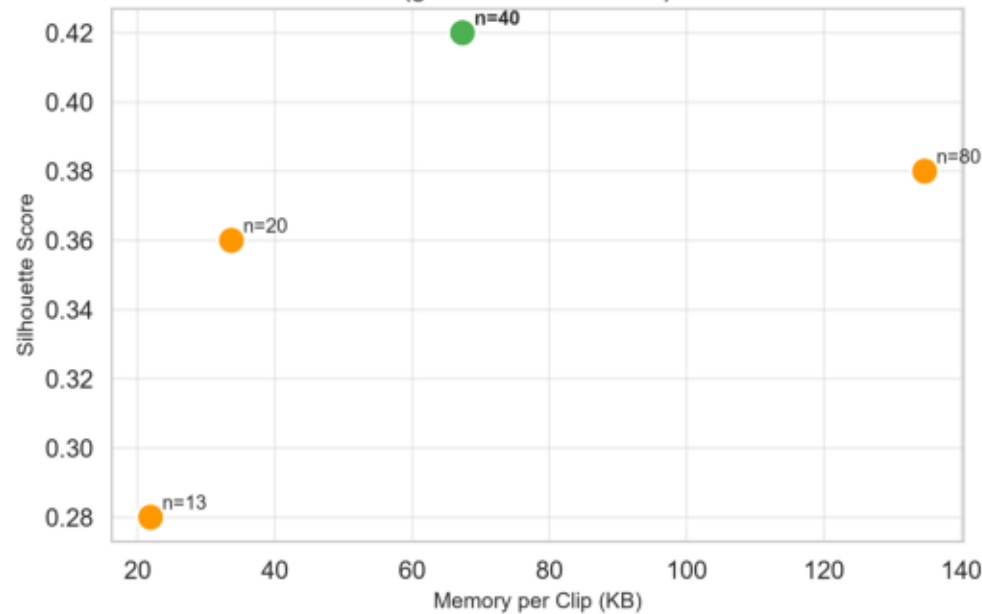
Recommended vs Baseline Configuration



Mel Resolution: Memory Cost
(green = recommended)



MFCC: Separability–Memory Frontier
(green = recommended)



Recommended config.yaml Update

```
# config.yaml – Week 3 Settings

feature_extraction:
  n_fft      : 2048
  hop_length : 512
  n_mels     : 128 # ← from study
  n_mfcc    : 40  # ← from study
  fmin      : 20.0
  cqt_bins  : 168
  cqt_bins_per_octave: 24 # ← from study

denoising:
  default_method: wavelet

training:
  batch_size   : 32
  learning_rate : 1.0e-3
  epochs      : 100
  random_seed  : 42
```


Week 3 Action Plan

WEEK 3 ACTION PLAN

P1 – Day 1

- Set `n_mels=128`, `n_mfcc=40`, `cqt_bins_per_octave=24`, `denoising.default_method=wavelet` in `config.yaml`
- Run full preprocessing pipeline on synthetic dataset
- Train CNN baseline (mel-only, no CQT branch)

P2 – Day 2–3

- Integrate CWRU Bearing Dataset (CWRULoader already implemented)
- Plug ExperimentTracker into training loop
- Implement Denoising Autoencoder (Week 4 baseline prep)

P3 – Day 4–5

- Run SNR robustness ablation (with / without denoising)
- Begin MIMII fan-recording integration
- Publish intermediate results to ExperimentTracker

P4 – Day 5

- Write Week 3 report (notebook 06)

SUCCESS CRITERIA

- CNN baseline F1-macro ≥ 0.85 on synthetic test set
- Training run fully logged in ExperimentTracker
- All splits stratified and group-by-source validated
- SNR robustness curve plotted